Continuous-Time Reinforcement Learning for Portfolio Optimization under Partial Information

A Review of Wu and Li (2024)

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The Pitch

Outline

1. Introduction & Motivation
2. Background
3. Methods: The RSEMV Problem & POEMV
4. Results
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Motivation

Why continuous-time RL?

- Financial markets evolve continuously, so discrete-time RL has poor motivation.
- Key challenge: explore without collapsing to a deterministic policy.

Why regime switching?

- Markets exhibit bull/bear non-stationarity, which is qualitatively important.
- EMV ignores regime structure.

Figure: S&P 500 (1992–2022): bull (green) / bear (red).

RL vs. the Classical Approach

Classical approach

1. Estimate model parameters from data.
2. Plug-in estimates into optimal control formula.

Issue: Mean return is hard to estimate — even small errors produce large changes in holdings, with no inherent self-correcting mechanism.

RL approach

1. Use filtering to obtain belief state p_t.
2. Learn optimal policy through repeated market interactions.

Advantage: Allows policy to self-correct, and estimation error impacts *only filtering*, not the policy directly.

Paper's Contribution

Wang and Zhou (2020) established the EMV framework for continuous-time mean–variance portfolio selection under full information. Wu and Li (2024) extend this to the partial-information regime-switching setting with the **POEMV algorithm**.

The MV Portfolio Problem

Over a fixed horizon $[0, T]$, we want to minimize terminal wealth variance subject to a mean target:

$$\min_u \ \mathrm{Var}[X_T^u] \quad \text{s.t.} \quad \mathbb{E}[X_T^u] = z, \quad u \text{ self-financed and admissible.}$$

Via a Lagrange multiplier ω, this is equivalent to:

$$\min_u \ \mathbb{E}\big[(X_T^u - \omega)^2\big] - (\omega - z)^2.$$

- **Time-inconsistent**: the variance term prevents a dynamically optimal solution.
- Wu and Li (2024) focus on the **precommitted strategy**, optimal only at $t = 0$.
- The multiplier ω^* is found via a **stochastic approximation**, with updates every $s = 10$ iterations to avoid destabilizing the policy learning loop.

Exploratory Stochastic Control

Key idea: replace point-action controls u_t with *distribution-valued* policies $\boldsymbol{\pi}_t \in \mathcal{P}(\mathcal{U})$.

The **exploratory wealth SDE** under stochastic policy $\boldsymbol{\pi}$:

$$d\tilde{X}_t^{\boldsymbol{\pi}} = \sigma\hat{\rho}_t\big(\int u\,\boldsymbol{\pi}(u)\,du\big)\,dt + \sigma\big(\int u\,\boldsymbol{\pi}(u)\,du\big)\,d\hat{B}_t + \sigma\sqrt{\int u^2\boldsymbol{\pi}(u)\,du - \big(\int u\,\boldsymbol{\pi}(u)\,du\big)^2}\,d\bar{B}_t$$

Entropy-regularized objective (temperature $\Lambda \geq 0$):

$$\min_{\boldsymbol{\pi}}\ \mathbb{E}\left[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda\int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot \mid t, X_t, p_t))\,dt\right] - (\omega - z)^2$$

- $\Lambda \to 0$ recovers deterministic optimal control.
- For LQ problems: optimal $\boldsymbol{\pi}^*$ is **Gaussian** with tractable mean and variance.
- **Regime switching** introduces a non-trivial $\mathrm{Cov}(\Delta X^{\pi}, \Delta p_t)$ term that couples wealth and belief dynamics.

Market Model & Wonham Filtering

Regime-switching Geometric Brownian Motion:

$$dS_t = S_t(\mu_t\, dt + \sigma\, dB_t), \quad \mu_t \in \{\mu_1, \mu_2\}$$

Regime $I_t \in \{1, 2\}$ follows a Continuous-time Markov chain with rates λ_1, λ_2.

Belief state: $p_t = P(I_t = 1 \mid \mathcal{F}_t^S)$
Wonham filter:

$$dp_t = \left[-(\lambda_1+\lambda_2)p_t+\lambda_2\right]dt+(\rho_1-\rho_2)p_t(1-p_t)\, d\hat{B}_t.$$

With Sharpe ratio $\rho_i = (\mu_i - r)/\sigma$, and

filtered Sharpe ratio $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$.

Separation principle:

- $p_t \in (0, 1)$ always — entrance boundaries, no boundary conditions needed.
- Partial-information problem $\Rightarrow$ fully observed HJB in (X_t, p_t).
- Wealth driven entirely by observable quantities:

$$dX_t = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$$

- Promotes p_t to a proper state and allows martingale theory to be applied directly.

Solution Structure (Theorems 1 & 2)

Minimizing over $\boldsymbol{\pi}$ in the HJB yields a **Gaussian optimal policy**. The ansatz

$$v = \frac{(x-\omega)^2}{f(t,p)} + g(t,p) - (\omega - z)^2$$

decouples the problem: f and g satisfy **linear parabolic PDEs** in (t,p) (no closed form, solved numerically). Let $\hat{\rho} = \rho_2 + (\rho_1 - \rho_2)p$:

$$f_t + \tfrac{1}{2}(\rho_1-\rho_2)^2 p^2 (1-p)^2 f_{pp} - \left[2\hat{\rho}(\rho_1-\rho_2)p(1-p) + (\lambda_1+\lambda_2)p - \lambda_2\right] f_p + \hat{\rho}^2 f = 0$$
$$g_t + \tfrac{1}{2}(\rho_1-\rho_2)^2 p^2 (1-p)^2 g_{pp} + \left[\lambda_2 - (\lambda_1+\lambda_2)p\right] g_p - \tfrac{\Lambda}{2} \ln \tfrac{\pi \Lambda f}{\sigma^2} = 0$$

- Degenerate at $p \in \{0,1\}$ — entrance boundaries mean no BCs needed.
- f decreasing in t; f_p, f_{pp} bounded — well-posedness via comparison principle.
- f encodes exploration variance; g encodes the entropy bonus.

Optimal Policy: Myopic–Hedging Decomposition (Theorem 2)

The optimal exploration policy is **Gaussian**:

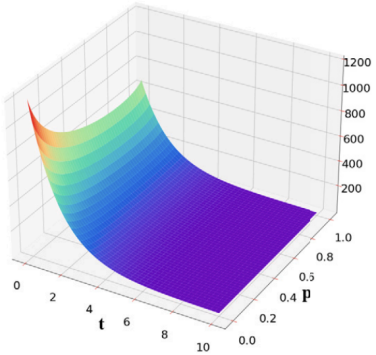
$$\boldsymbol{\pi}^*(\cdot \mid t, x, p) = \mathcal{N}\left(m^*(t,x,p),\ \frac{f(t,p)\Lambda}{2\sigma^2}\right)$$

where the **mean decomposes** as:

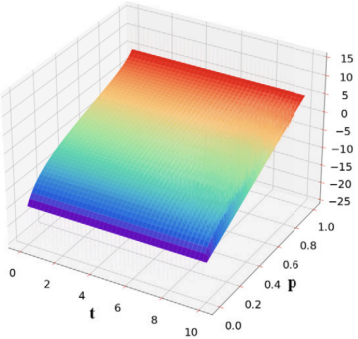
$$m^*(t,x,p) = \underbrace{-\frac{x-\omega}{\sigma}\left[\rho_2 + (\rho_1-\rho_2)p\right]}_{\text{myopic: current filtered Sharpe ratio}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1-\rho_2)p(1-p)\frac{f_p}{f}}_{\text{hedging: regime transition risk}}$$

- Hedging demand is **largest at** $p_t \approx \frac{1}{2}$ (maximum uncertainty), reducing risky exposure; vanishes when beliefs concentrate near 0 or 1.
- Setting $\rho_1 = \rho_2$ eliminates hedging and recovers the single-regime case.
- Policy variance $\propto \Lambda f(t,p)$ decays as $t \to T$, so exploration naturally winds down as we approach stopping time.

PDE Surfaces: $f(t,p)$ and $\gamma^*(t,p)$



(a) $f(t,p)$



(b) $\gamma^*(t,p)$

- $f(t,p)$ **decreases in** t: exploration variance decays over the horizon.
- $\gamma^*(t,p)$ **increases in** p: invest more in the risky asset as the bull-market probability rises (when $x < \omega$).

The POEMV Algorithm

Parametrize $\ln f^\theta$ and g^θ as order-m polynomials in p and $(T-t)$ (at $m=2$: MAE $= 1.03\%$ for $\ln f$, 0.06% for g). Policy:

$$\boldsymbol{\pi}^\phi(\cdot \mid t,x,p) = \mathcal{N}\left(-(x-\omega)\left[\phi_1\left(p - \frac{f_p^\theta}{f^\theta}p(1-p)\right) + \phi_2\right],\ \Lambda e^{\phi_3} f^\theta(t,p)\right)$$

Each episode:

1. **Filter:** compute p_{t_k} via discretized Wonham filter.
2. **Critic (PE):** gradient descent on θ minimizing the martingale loss — any nonzero drift in $V^\theta - \Lambda \int \mathcal{H}\, ds$ signals misspecification.
3. **Actor (PG):** score-function gradient step on ϕ.
4. **Lagrange update** (every 10 episodes): $\omega \leftarrow \omega - \eta_\omega(\bar{X}_T - z)$.

Polynomial parametrization outperforms Neural methods like DDPG/SAC; with fewer parameters, we have more stable training.

Simulation Setup: Calibrated Parameters

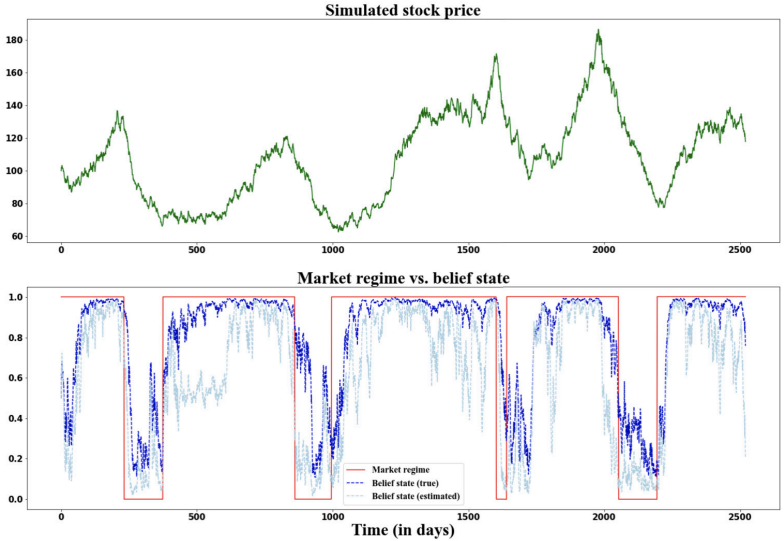
Parameters calibrated from 20 years of daily S&P 500 data (Dai et al. 2010 heuristic):

μ_1	μ_2	σ	λ_1	λ_2
0.25	-0.73	0.22	0.36	2.89

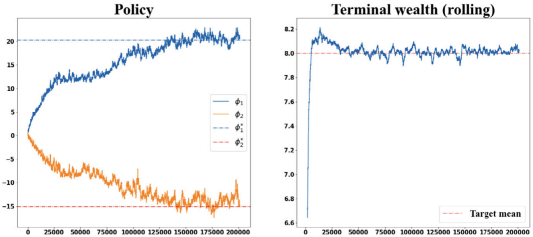
Table: True environment parameters (Table 1 of Wu and Li (2024))

- Horizon: 10 years; $x_0 = 1$, target $z = 8$, $\Delta t = 1/252$, $\Lambda = 1$; 200,000 training episodes.
- **EM algorithm** produces implausible transition rates ($\hat{\lambda}_1 \approx 128$, $\hat{\lambda}_2 \approx 132$) due to low SNR at daily frequency, so a heuristic approach is adopted.

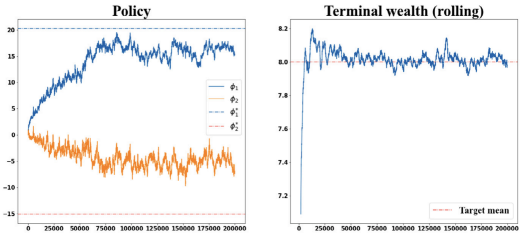
Filtering Quality: Belief State Tracking



Simulation Convergence Results



(a) POEMV with true environment parameters



(b) POEMV with estimated environment parameters

Simulation: Out-of-Sample Performance

Out-of-sample evaluation on 100,000 independent 10-year test paths:

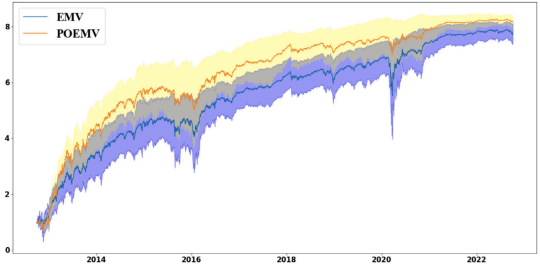
Table: Out-of-sample performance (Table 3 of Wu and Li (2024))

	ϕ_1	ϕ_2	Mean	Variance	Sharpe
Optimal (FD)	20.25	-15.08	7.99	0.067	27.00
POEMV (true)	20.56	-15.05	7.99	0.083	24.26
POEMV (est.)	15.23	-7.29	8.03	0.170	17.05

- With true parameters, POEMV nearly matches the finite-difference optimal.
- With estimated parameters, variance (0.170) remains far smaller than classical alternatives.
- **Error appears driven by filtering error**, not policy-based learning error.

Empirical Results: Wealth Trajectories

30-year S&P 500 data; 20-year train / 10-year test. 30 $\beta \approx 1$ stocks as index proxies (≈75k episodes). **EMV** same architecture, no regime awareness — isolates regime contribution.



(a) Daily rebalancing

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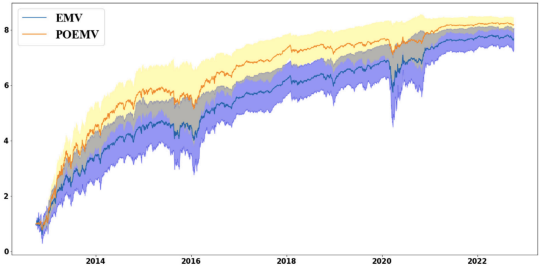

(b) Monthly rebalancing

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POEMV: faster bull-market growth, better protection in **March 2020**.

Empirical Results: Performance Tables

(a) Daily rebalancing

	Mean	Var	Sharpe
POEMV	8.19	**0.224**	**15.38**
EMV	7.79	0.453	10.09
Classical	8.15	1.537	5.77

(b) Monthly rebalancing

	Mean	Var	Sharpe
POEMV	8.19	**0.284**	**13.49**
EMV	7.69	0.597	8.66
Classical	8.22	10.081	2.27

- POEMV variance $\approx \frac{1}{2}$ of EMV (daily); gap $> 10\times$ vs. classical (monthly).
- POEMV also outperforms under leverage constraints $\ell = 2, 3$ (Table 5).

Open Problems

Wu and Li (2024) establish structural results (Theorems 1–2) but **no end-to-end convergence guarantee** exists. Four challenges remain:

1. **Convergence:** Filter error, polynomial approximation error, and stochastic gradient error are not jointly analyzed. A rigorous decomposition and comparison of errors remains open.
2. **Time consistency:** The separation principle holds for the *precommitted* strategy only. Extending to *equilibrium* strategies requires new theory.
3. **Joint estimation & control:** The current architecture *sequences* filtering and learning. A fully model-free coupling of the Wonham filter and actor–critic updates remains open.
4. **Scalability:** Two regimes, one risky asset. Multi-asset portfolios and richer latent structure interact non-trivially with the filtering and control problems.

Conclusion

1. **Gaussian optimality survives partial information** via the Wonham filter and separation principle — the foundational result enabling POEMV.
2. **The martingale actor–critic transfers cleanly** from full to partial information with minimal modification.
3. **Regime-awareness materially improves performance:** POEMV halves out-of-sample variance vs. EMV, while maintaining mean wealth target.
4. **Filtering error, not learning error, appears to be the dominant bottleneck**, pointing toward joint parameter estimation and control as the most impactful open problem.

References I

- Wang, Haoran and Xun Yu Zhou (2020). "Continuous-Time Mean–Variance Portfolio Selection: A Reinforcement Learning Framework". In: *Mathematical Finance* 30.4, pp. 1273–1308. DOI: 10.1111/mafi.12281.
- Wu, Bo and Lingfei Li (2024). "Reinforcement Learning for Continuous-Time Mean-Variance Portfolio Selection in a Regime-Switching Market". In: *Journal of Economic Dynamics and Control* 158, p. 104787. DOI: 10.1016/j.jedc.2023.104787.

Appendix: POEMV Algorithm (Algorithm 1)

Require: learning rates $\eta_\theta, \eta_\phi, \eta_\omega$; wealth x_0; target z; horizon T; $\Delta t = T/K$; Λ; iterations M; Lagrange update frequency s.

Initialize: Estimate $\tilde{\sigma}, \tilde{\mu}_i, \tilde{\lambda}_i$ from historical data. Set $\theta, \phi = \mathbf{0}$.

For $m = 1$ to M:

1. **Episode:** At each t_k, observe (S_{t_k}, X_{t_k}); filter p_{t_k} via Wonham; sample $u_{t_k} \sim \boldsymbol{\pi}^\phi$.
2. **PE update:** $\theta \leftarrow \theta + \eta_\theta \, \Delta\theta$ (martingale loss gradient).
3. **PG update:** $\phi \leftarrow \phi - \eta_\phi \, \Delta\phi$ (score-function gradient).
4. **If** $(m \bmod s) = 0$: $\omega \leftarrow \omega - \eta_\omega \left(\frac{1}{s} \sum X_T^j - z \right)$.

Appendix: Discretized Wonham Filter

Apply Itô's formula and Euler scheme to $q_t^1 = \log p_t$ and $q_t^2 = \log(1 - p_t)$ to guarantee $p_{t_k} \in (0, 1)$:

$$q_{t_k}^1 = q_{t_{k-1}}^1 + \left[\tilde{\lambda}_2 e^{-q_{t_{k-1}}^1} - (\tilde{\lambda}_1 + \tilde{\lambda}_2) - \tfrac{1}{2}(\tilde{\rho}_1 - \tilde{\rho}_2)^2 (1 - e^{q_{t_{k-1}}^1})^2\right] \Delta t + (\tilde{\rho}_1 - \tilde{\rho}_2)(1 - e^{q_{t_{k-1}}^1}) \Delta \hat{B}_{t_k}$$

Then: $p_{t_k} = e^{q_{t_k}^1} / (e^{q_{t_k}^1} + e^{q_{t_k}^2})$, guaranteed in $(0, 1)$.

Innovation process increment:

$$\Delta \hat{B}_{t_k} = \frac{1}{\tilde{\sigma}} \left[\log \frac{S_{t_k}}{S_{t_{k-1}}} - (\hat{\mu}_{t_{k-1}} - \tilde{\sigma}^2/2)\Delta t\right], \quad \hat{\mu}_{t_{k-1}} = \tilde{\mu}_1 p_{t_{k-1}} + \tilde{\mu}_2 (1 - p_{t_{k-1}})$$

Appendix: Estimation Method Comparison (Table 2)

Table: Heuristic vs. EM algorithm (100 simulated paths; Table 2 of Wu and Li (2024))

	μ_1	μ_2	λ_1	λ_2	σ
True	0.25	−0.73	0.36	2.89	0.22
Heuristic	0.58 (0.10)	−0.69 (0.13)	1.21 (0.42)	2.09 (0.59)	0.22 (0.003)
EM	0.51 (0.65)	−0.21 (0.28)	128.59 (90.14)	132.21 (88.03)	0.25 (0.12)

- EM fails on transition rates — low SNR at daily frequency causes it to identify regime switches from noise.
- Heuristic adopted throughout; both methods estimate σ accurately.

Appendix: Leverage Constraint Results (Table 5)

With leverage cap ℓ and target $z = 4$ (daily rebalancing):

Table: Performance under leverage constraints (Table 5 of Wu and Li (2024))

	$\ell = 2$		$\ell = 3$	
	Variance	Sharpe	Variance	Sharpe
POEMV	**0.690**	**3.23**	**0.691**	**3.39**
EMV	0.766	2.91	0.734	3.10

POEMV outperforms EMV on mean, variance, and Sharpe in both cases — regime-awareness is beneficial even under leverage constraints.